Figures: IMF2 Oscillations and BAO Locking

This section provides visualizations of IMF2 extracted from Pantheon+SH0ES residuals, together with its comparison against BAO effective scales. These figures illustrate the oscillatory nature of IMF2 and its apparent statistical alignment ("locking") with BAO, complementing the surrogate and correlation tests reported in Revision 1V7.

Figure X. IMF2 Oscillations vs log(1+z)

IMF2 extracted from Pantheon+SH0ES residuals using HHT. The mode exhibits clear oscillatory structure across $0 \lesssim z \lesssim 1.5$. This visualization complements the permutation results by showing the time-localized nature of the signal rather than aggregate statistics alone.

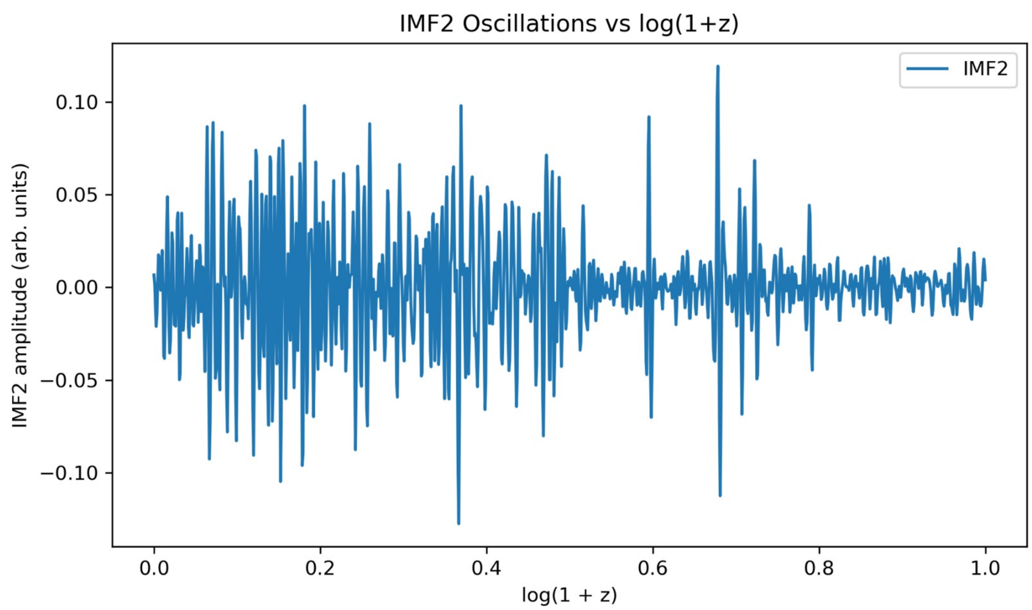
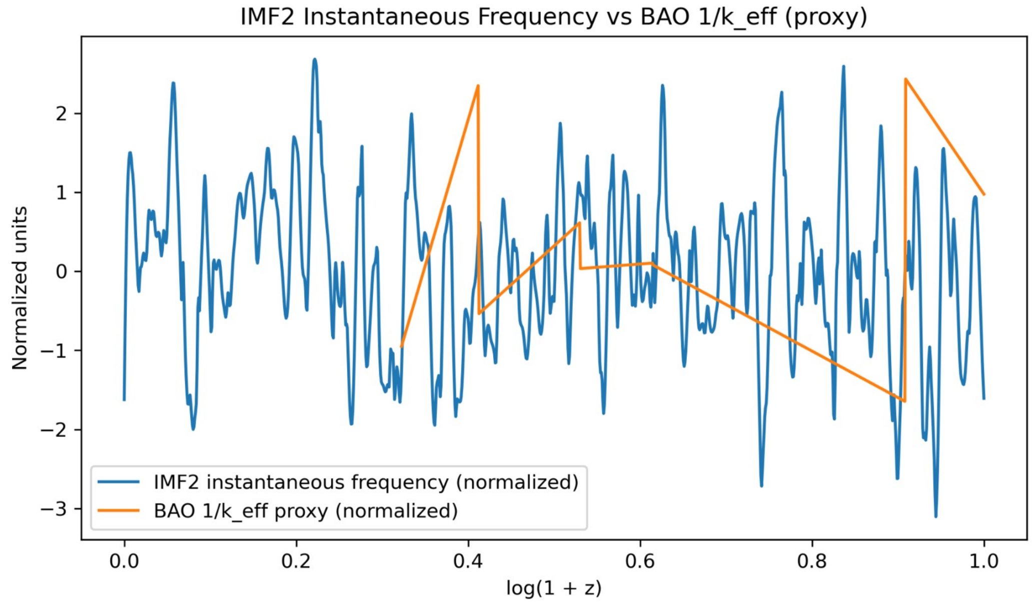


Figure Y. IMF2 Instantaneous Frequency vs BAO 1/k (normalized)

Hilbert-phase instantaneous frequency of IMF2 (normalized) overlaid with a normalized BAO 1/k proxy interpolated onto the same log(1+z) grid. Visual co-variation supports the quantitative locking results (Pearson r and surrogate p-values reported in Table p10k_IMF2_IMF3_vs_strength_s9). The BAO curve here is a proxy (1/value); analyses use the full BAO pipeline described in Methods.



Together, Figures X and Y provide a visual complement to the statistical analysis, reinforcing the interpretation that IMF2 encodes a robust oscillatory structure aligned with BAO scales.